

Building Bridges

Engineering

Ages 3+



Set-Up

Add some plastic or paper cups and cardboard tubes to the tray. Place a selection of lolly sticks on the tray, alongside straws and a couple of rolls of tape. Provide a selection of small world toys, such as cars, trees and people (or you may wish children to choose their props themselves).

You will need:

- Plastic or paper cups
- Cardboard tubes (kitchen roll tubes)
- Straws
- Tape
- Large and small lolly sticks
- Small world toys: cars, people, animals and trees

Investigate and Play

Independent Exploration

Children can inspect the items on the tray and investigate how they can build bridges to include in a small world scene. They may wish to build bridges for cars, people or animals. Children may explore making bridges of different heights and lengths using the cardboard tubes, cups, lolly sticks and straw. They may decide to use the tape to stick the straws together to make a walkway for their bridge.



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While children are exploring the play tray, supporting adults, if available, can use these prompts to encourage high-quality interactions.

Predict and Plan

Children may:

- predict how many lolly sticks they need to add to the cups to hold their small word toys;
- think about how far apart the cups/tubes should be to make a strong bridge;
- plan how to use the cardboard tubes, straws and tape to make longer bridges.

Possible Adult Prompt:

Invite children to think about how they will use the cups, tubes, straws and lolly sticks to make their bridges strong enough for the toys to cross.

Test and Observe

Children may:

- test the strength of the different bridges by placing small word toys on top to see if they hold;
- adjust their bridge by adding more straws or lolly sticks or moving the cups and tubes;
- notice the different heights and strengths of the bridges in the tray.

Possible Adult Prompt:

Encourage children to describe what happens to their bridge when they move the distance of the cups or tube, using words such as 'stronger', 'weaker', 'balanced', 'steady' or 'wobbly'.

Analyse and Evaluate

Children may:

- talk about which bridges were the strongest and weakest and why;
- explain what they found out, for example, 'the bridge broke because the cups were too far apart';
- discuss alternative ways that they might make bridges next time.

Possible Adult Prompt:

Invite children to evaluate what they would do differently next time to make their bridges even stronger. Encourage them to consider whether they would use different materials and why.

Key Vocabulary

Bridge, strong, weak, steady, wobbly, build, support, balance, tall, long, short, heavy, light.

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